SECONDSMATTER

Project Proposal - Capstone Project 2026/2027

1. Team

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Member	Student ID	Email	Role
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2. The idea

Problem and users - In Lisbon, emergency resources are limited and several urgent calls can arrive at the same time. The closest ambulance is not always the best one to send if that decision leaves another area with no nearby coverage. SecondsMatter is a decision-support application for emergency dispatch and coordination teams.

What already exists and what is different - Lisbon has already tested predictive emergency planning through projects such as LxAnalyticsHub, and previous research has studied ambulance demand forecasting and optimisation. Our focus is the operator's decision in two different moments: a planning mode for analysing coverage and relocating resources, and a live dispatch mode that becomes much simpler when a new emergency arrives and a decision must be made in seconds.

What users can do - In planning mode, the user sees demand by area, current coverage and possible relocations on a map. In live dispatch mode, the interface shows only the few relevant options - suitable available units, estimated arrival time and the effect of sending each unit on the remaining coverage. Two operational indicators will support these comparisons: estimated response time and coverage rate. The system recommends, helping a person make decisions.

Data we can use - For development and testing, we can use the public NYC EMS Incident Dispatch Data, which contains incident-level timestamps, call type and severity, dispatch/response information and geographic areas. This will let us test the forecasting method and measure prediction error by area and time period. For the Lisbon prototype, we can use public INEM/SNS data on daily emergency occurrences by priority and activations by type of emergency resource, Lisbon geographic and traffic data from the municipality, and weather data from IPMA. The Lisbon version will be recalibrated using Portuguese demand statistics and Lisbon context rather than assuming that New York patterns transfer directly.

Data we do not have publicly. We do not currently have incident-level INEM records for Lisbon (important variables). Where these variables are needed, we will use calibrated synthetic operational data and separate it clearly from real Portuguese data. If anonymised academic data becomes available from INEM, it can replace part of the simulated layer.

Role of the conversational model - Forecasting, travel-time calculations and optimization are handled by our own models and functions. The conversational component interprets free-text questions such as "what happens if unit 12 becomes unavailable?", calls the appropriate functions and explains the returned options in plain language. It does not assign clinical priority or dispatch a resource without operator confirmation.

3. One interaction, end to end

A high-priority emergency is received in Alvalade and the operator asks, "Which unit should I send?" The model receives the incident information and the current resource status, identifies that a dispatch comparison is needed and calls our ranking and coverage functions. Our code compares suitable available resources using estimated travel time and the effect each choice would have on coverage in the rest of Lisbon. The interface returns two or three clear options; the operator confirms one, and the system immediately updates that resource as unavailable and recalculates the remaining coverage.

4. Questions

- Will the course cover spatial/map analysis, for example working with parishes, operational zones or geographic grids for demand and coverage?
- Given that these decisions can affect lives and may need to be made in seconds, should the system remain strictly a recommendation tool, or is there any acceptable level of automated action in the project?